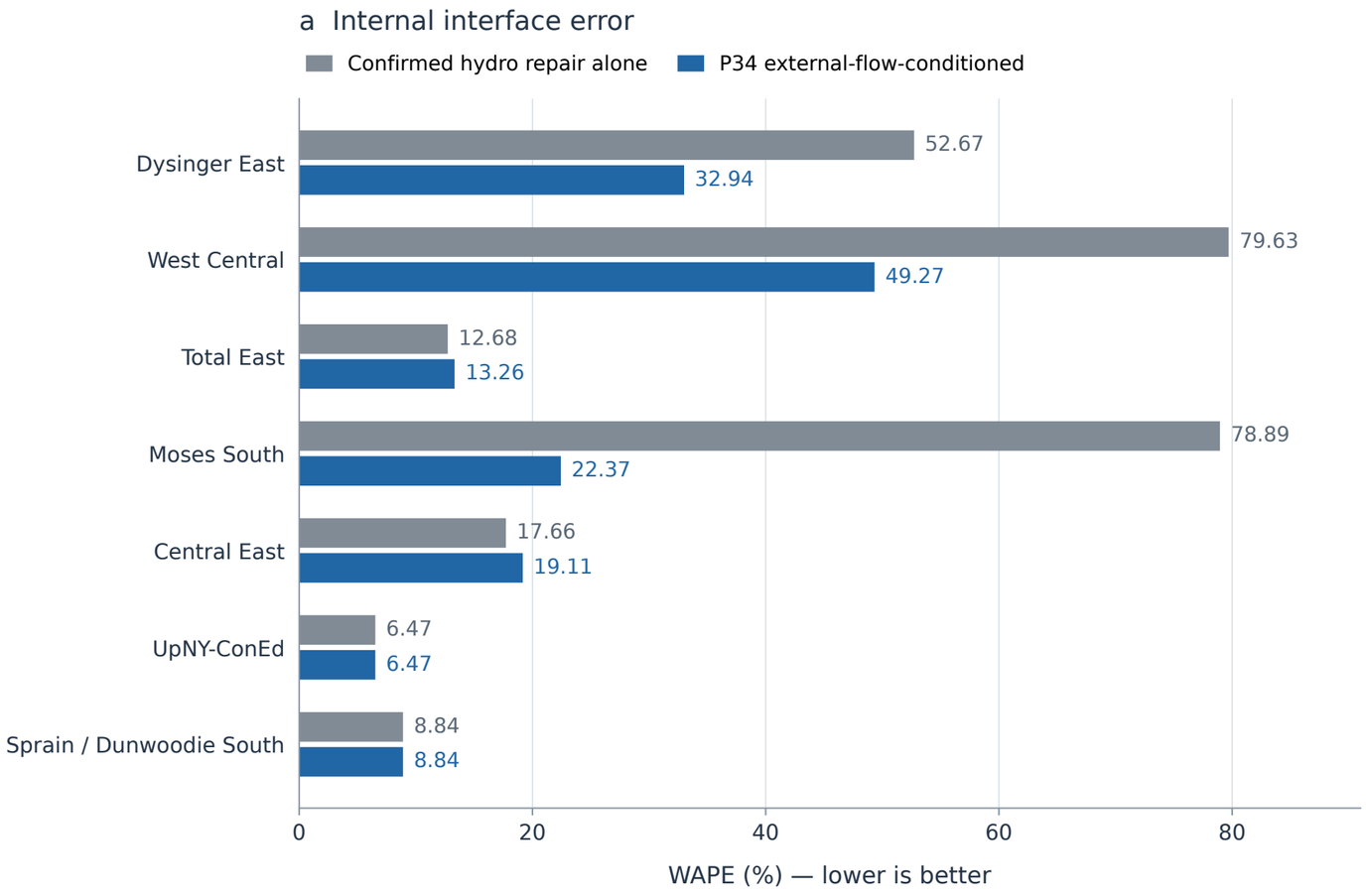
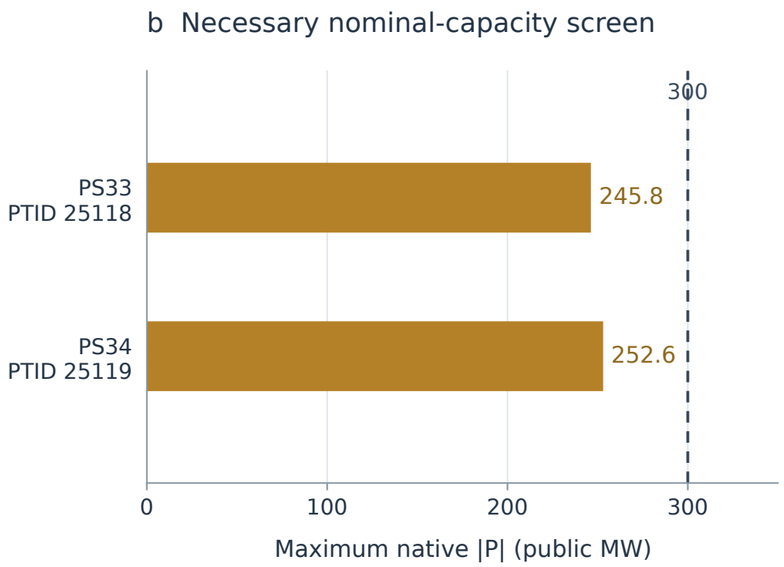
Verified St. Lawrence PAR flow inputs: seven-interface reconstruction

2025 dated topology · panel a: the same 8,663 qualified hours · panel b: native-resolution source samples





105,998 raw P34 samples per device; includes Jan 1, 2026 endpoint day. Screen applied before hourly averaging.

300 MVA is nominal apparent capacity. |P| ≤ 300 MW is only a necessary outer screen, not an operating P-limit.

P34 observed external flows are inputs; no internal interface flows were used to fit the model or choose direction.

Hardware tap trajectories, phase-dependent thermal limits, P-Q/current capability and contingency feasibility remain unverified.

WAPE = 100 × Σ|modeled − observed| / Σ|observed|. Total East and Central East WAPE increase after conditioning.

Direction evidence: NYISO Par Flow Direction Diagrams, page 7; legend page 26 (effective December 3, 2024).

Sources: comparison/interface_summary.csv; controls/necessary_p_only_screen.json. Exact plotted rows and source SHA-256 hashes accompany this figure.